MIDTERM PROJECT PROPOSAL

Outdoor Adventure Planner

02/22/2026

PREPARED FOR

SE/COM S 3190 - Construction of User Interfaces

Iowa State University
Department of Computer Science

PREPARED BY
Peter Yehl

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1 Introduction

I, Peter Yehl am from team MS_6. I am a senior majoring in Computer Engineering. I have experience creating my personalized portfolio using HTML & CSS. I am taking this class to deepen my understanding of frontend frameworks, syntax, and semantics. This project will help to accomplish this by allowing me to work on a project I enjoy while still teaching me the workflow of frontend development.

2 Purpose of the proposal

I aim to develop a website that is both helpful and visually appealing for outdoor lovers. Through the website “Outdoor Adventure Planner,” I plan to provide users with a seamless experience when planning trips to national parks and other outdoor areas. I believe being outside is essential for people’s lives. Therefore, I want to create an interactive website that allows users to easily plan trips, helping them to be prepared.

3 Goals and Objectives

My project’s goals and objectives include:

- Developing a user-friendly website with an attractive design.
- Providing information about filtered outdoor areas that provide users with JSON information about nearby trails, camping areas, and climbing routes.
- Creating an easy-to-navigate interface for users to explore the difficulty and length of trails, information about the amenities of campsites, and climbing routes and their difficulty.
- Demonstrating proficiency in HTML, CSS, and JavaScript for web development.
- Incorporate a search feature that takes input for a particular location (accepting town names and states, or national parks) with a dedicated card with more information about the area that includes either the trail, camping, or climbing information.
- One of the primary objectives is to consistently update the website to provide the latest information as new trails are formed, campsites are developed, or climbing routes are discovered.
- Incorporate a personalized page about the author that also includes a window about the terms and conditions.

4 Project

With the project "*Outdoor Adventure Planner,*" I intend to create a website for the outdoors. I believe that being outside is the greatest place to be. Engaging in activities outside offers a multitude of health benefits and allows the individual to discover new forms of personal expression. My aim is to present the rich information that national parks contain and to allow users to input their own trails, climbing routes, and campsites they've discovered.

Our project will feature several key components:

- A search bar for finding specific locations, filtering to allow users to discover only certain items in an area.
- Interactive map icons of trails, campsites, and climbing routes, allowing users to view more information about their chosen item.
- A page for terms and conditions, FAQ, and creator information.
- A navigation bar, allowing users to access all pages quickly and seamlessly.
- A my trips page, letting users save their desired destinations in a specific collection that is showcased alongside other trips.

4.1 Example Wireframe for Homepage:

- A welcoming interface with navigation to trails, campsites, climbing, and the FAQ page.
- Backdrop images on each of the buttons will allow users to understand more of what's behind each page.
- Search bar will allow users to enter a map page, allowing users to filter or add locations they want to a specified list for their particular trip.
- The plan a trip button will take users to a chronologically organized section, made to entice users to focus on particular parts of their trip first.

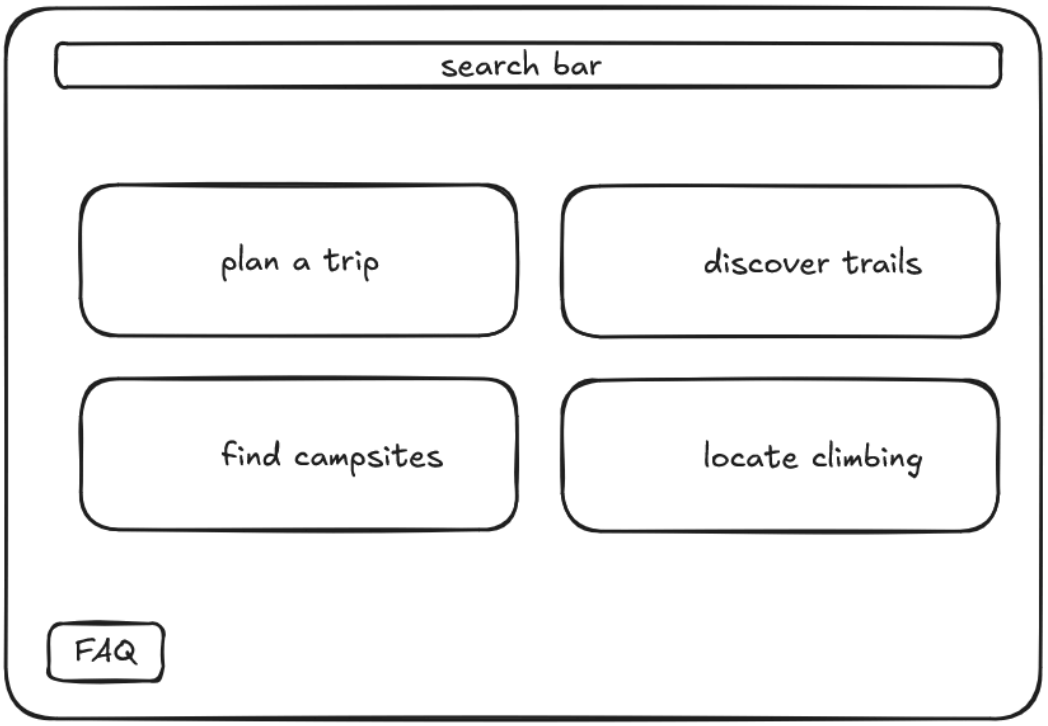

Figure 1: Home Page Wireframe - Designed on Excalidraw

4.2 Example Wireframe for Trips Page:

• A grid layout displaying a user’s planned trips.

• Each item should have a hover effect, enlarging additional brief details about the trip.

• The user should be able to interact with the edit or view trips buttons, allowing them to see the trip and potentially make changes to it.

• The search for my trips bar allows users to locate the trip they're looking for based on the keywords in the title

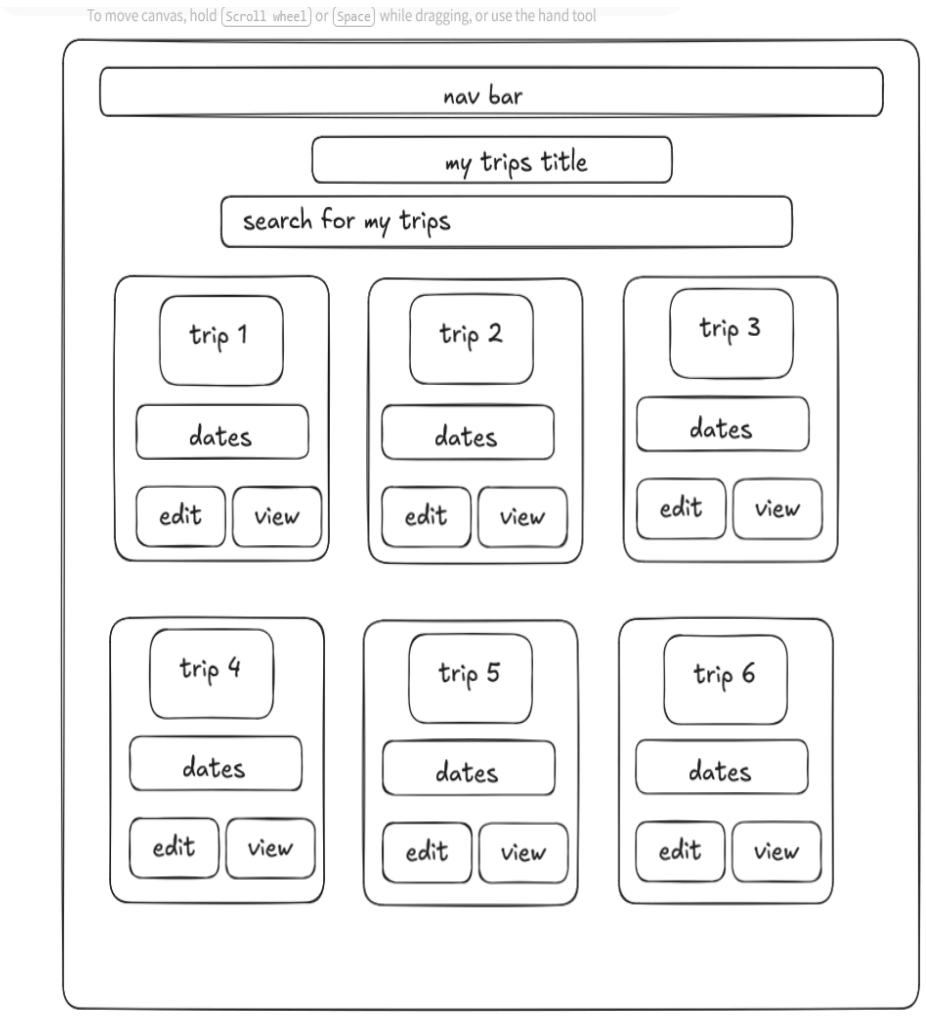

Figure 2: Trips Page Wireframe - Designed on Excalidraw

4.3 Description of Images and Figures:

- The homepage wireframe will feature an organized layout that allows for easy navigation to all other parts of the website.
- The trips page wireframe showcases the interactive features of the website. Allowing for users to understand the many choices they have when navigating. Lastly, the page is shown to be visually enticing, letting users explore multiple areas of the website while being pleased with the design.

5 Resources

1. Web development programming languages like HTML, CSS, and

JavaScript.

2. Use of JSON files to store information about different trails,

campsites, and climbing routes.

3. Image assets from Google for different national parks.

4. Use of DOM to include HTML & CSS code with/ the JavaScript file.

5. Time allocation for me to collaborate on the project.

8. Use GITLAB as my version control remote repository to push and pull my work to keep my work up to date.

9. Leverage GitLab issue boards to distribute tasks among myself, improving the development experience, and integrate the SCRUM model into our workflow.

11. Use of CDN from Bootstrap can enhance the loading speed of our website by distributing assets across multiple servers geographically.

12. Use DrawIO to design the prototype for different pages of the website.

13. My objective is to elevate the functionality of our website through the integration of JSON data created by me.

14. Use Excalidraw to create wireframes for the main web pages, including the homepage, camping areas, trails, create page, and climbing routes. These wireframes will guide the development of my user interface, ensuring a user-friendly experience.

6 Future work

The Outdoor Adventure Planner is designed to facilitate easy expansion for future features. Features I plan to incorporate after the initial pilot version is launched include:

- Integrate a friends feature to allow for friends to be invited on trips
- Implement a partition section of Google Maps street view that allows users to visually place themselves at a hike, campsite, and climb of their choosing
- Offer a chat feature allowing users to interact with one another through direct messaging
- Explore advertisement and monetizing opportunities.

This project lays the groundwork for my final project and may incorporate more advanced features, such as Google Maps Street View for location data.

7 Final Comments

This is my proposal for The Outdoor Adventure Planner. I believe this project has a lot of potential with the number of features that can be scaled. I also believe this project will significantly enhance my understanding of frontend software development, as it will involve many programming languages working together to bring a product to life. I aim to give users a seamless experience that helps them explore the outdoors. Thank you very much for considering my Outdoor Adventure Planner proposal. I am open to feedback and constructive criticism. You can reach me at: pmyehl@iastate.edu